

Document Q&A Assistant (Mini RAG)

Goal

The objective of this project is to build a **Document Question Answering Assistant** using a **Retrieval-Augmented Generation (RAG)** approach. The system allows users to upload a document and ask questions, with answers generated based on the document content.

◆ System Overview

This project follows a RAG pipeline where the document is processed, embedded into a vector database, and queried using a language model to generate accurate answers.

Flow Design

Upload → **Embed** → **Retrieve** → **Answer**

◆ Implementation Steps

1. Document Loader

- Accepts input documents (PDF, TXT, etc.)
- Extracts raw text from the uploaded file

2. Text Splitter

- Splits the document into smaller chunks
- Ensures efficient embedding and retrieval
- Helps maintain context within manageable limits

3. Vector Store

- Converts text chunks into embeddings
- Stores them in a vector database
- Enables similarity-based search

4. Retriever

- Fetches the most relevant chunks based on user query
- Uses similarity search (e.g., cosine similarity)

5. LLM Integration

- Connects retrieved chunks to a language model
- Generates answers using both query + retrieved context

6. Query Answering

- User inputs a question

- System retrieves relevant data
- LLM generates a final response

Bonus Activities Done:

Knowledge Base Integration

- Upload multiple documents or FAQs
- Build a reusable knowledge system
- Improve answer coverage

Hybrid Bot (Rules + AI)

- Combine rule-based responses for common queries
- Use AI fallback for complex or unknown questions

Conclusion

This project demonstrates the practical implementation of a **Mini RAG system**, combining document retrieval with generative AI. It highlights how unstructured data can be transformed into an interactive question-answering system, forming the foundation for advanced AI assistants.